Feature Specification: Generic Work Center Request

Feature Branch: `015-generic-work-center-request`
Created: 2026-10-27
Status: Draft
Input: Owner request: "A Work Center that is not reliant on a Work Order. Instead of pulling request data from the work order the requester is responsible for stating what they need. Part Numbers must be validated against part numbers in Visual. Dunnage uses the Substitute as a page. If picking up scrap the requester is responsible for stating the scrap type during the request process. A new Generic Request Flow and Generic Request cards need to be established as new modules/projects."

The decisions below were settled with the owner in review on 2026-10-27: the generic designation is a flag on the work center catalog; a stated part is proved real against the WIP application or Infor Visual according to a WIP/finished-good choice the requester makes first; the dunnage step *is* the substitute catalogue, shown as a page; scrap types come from the configured lugger arrangement; generic requests share the existing queue and board but draw their own card; the flow and its views are built as new projects; and a die is found through a new Visual-backed search by die number or part number that lists every die a part has, with locations.

User Scenarios & Testing

User Story 1 - A work center with no work order can still ask for what it needs (Priority: P1)

A press operator works at a work center that has nothing on the board — no active setup job, no work order to point at. Until now the request wizard stopped at that work center because it had nothing to read. The operator opens the wizard, picks that work center, and instead of the app reading a job, the app asks them what they need: which category and item, which part, and the few questions that item requires. They confirm, and the request appears on the same waitlist board as every other request, ready for a handler.

Why this priority: this is the feature. Every other story exists to make one of those answers provable or to draw the result correctly. Without it, a work center can only ask for material by first inventing a work order.

Independent Test: mark one work center as generic, raise a request at it with nothing but a stated part, and confirm the request lands on the board with no work order anywhere on it, while a non-generic work center still follows the work-order flow unchanged.

Acceptance Scenarios:

1. **Given** a work center is designated generic, **When** the requester selects it on the work center step, **Then** the wizard continues without an active job being required and never refuses the request for a missing job.
2. **Given** a generic work center is selected, **When** the wizard continues, **Then** the Category step offers the same categories a work-order request offers, and the Item step offers every item a work-order request offers.
3. **Given** a generic work center is selected, **When** the wizard reaches any step that a work-order request fills from the job (die, component, coil, hopper, dunnage, scrap), **Then** the requester answers that step themselves and no job value is read, shown, or guessed.
4. **Given** a requester has answered everything the chosen item requires, **When** they confirm, **Then** one request is raised into the waitlist queue and appears on the board.
5. **Given** a work center that is **not** designated generic, **When** it is selected, **Then** the existing work-order flow runs exactly as it does today.
6. **Given** a generic work center that happens to have an active job, **When** a request is raised there, **Then** the generic flow is used and the job is not read.

User Story 2 - A stated part is proved to be a real part (Priority: P1)

A machinist needs a part brought to their work center. They are asked one question before anything else: is the part they want a WIP part or a finished good? They answer, then type the number into a box that completes as they type and only ever offers genuine parts. A WIP choice is proved against the WIP floor's records; a finished good is proved against Infor Visual. Once the part is proved, the app says whether any of it is on hand — and a part with nothing on hand is still allowed through, because the machinist may know something the stock figure does not.

Why this priority: it is the only thing standing between a free-text part number and a handler walking to a bin that does not exist. Everything downstream — the card, the pick, the audit — assumes the number is real.

Independent Test: raise one request for a WIP part and one for a finished good; then try to submit a made-up number, and a real number with no stock, and confirm the first is refused while the second

warns and proceeds.

Acceptance Scenarios:

1. **Given** the part question is shown, **When** it appears, **Then** it asks first whether the part is WIP or a finished good, and a part number cannot be accepted before that answer is given.
2. **Given** WIP is chosen, **When** the requester types a number, **Then** completions are drawn from the WIP application and the accepted number must exist there.
3. **Given** a finished good is chosen, **When** the requester types a number, **Then** completions are drawn from Infor Visual and the accepted number must exist there.
4. **Given** a typed number that matches no genuine part, **When** the requester tries to continue, **Then** the flow refuses to advance and says the number is not a known part.
5. **Given** a proved part, **When** the flow checks availability, **Then** it reports whether any of that part is on hand and which system answered.
6. **Given** a proved part with no stock on hand, **When** the requester continues, **Then** they may still raise the request, having seen a warning that none was on hand.
7. **Given** the WIP application or Infor Visual cannot be reached, **When** the requester tries to prove a part, **Then** no unverified number is accepted, the step reports that source as unavailable with the time of the attempt and a retry, and the flow does not fall back to any cached or sample part list.
8. **Given** a part with no stock, **When** the request is raised, **Then** the stored request carries the part and the fact that none was on hand — never an invented quantity or an invented stock figure.
9. **Given** the requester changes the answer from WIP to finished good, or back, **When** the change is made, **Then** the previously typed number is cleared and must be proved again against the newly chosen source.

User Story 3 - Dunnage is chosen from the substitute catalogue, as a page (Priority: P1)

A material handler needs dunnage for an upcoming run, and the app cannot read a job to tell them which dunnage the run carries. The dunnage step opens the same dunnage catalogue that the substitute picker shows today — the image-backed list of parts — but as a page of the wizard rather than a dialog laid over it. They pick the part they need and carry on, one dunnage part per request.

Why this priority: without a page, the generic flow has no way to name dunnage at all, and dunnage is one of the request categories the plant raises every shift.

Independent Test: raise a generic dunnage request by choosing a part from the catalogue page, and confirm the chosen part appears on the confirmation step and on the card.

Acceptance Scenarios:

1. **Given** the dunnage item is chosen on a generic request, **When** the dunnage step is entered, **Then** the dunnage catalogue opens as a full page of the wizard.
2. **Given** the catalogue page, **When** the requester picks a part, **Then** that part becomes the request's dunnage and the wizard offers the next step.
3. **Given** the catalogue page, **When** the requester leaves it without choosing a part, **Then** no dunnage part is recorded and the confirmation step lists none.
4. **Given** a generic dunnage request, **Then** exactly one dunnage part is carried, and a second part is a second request.
5. **Given** the catalogue cannot be read, **When** the page opens, **Then** it says the catalogue is unavailable with a retry, and offers no invented or sample parts.

User Story 4 - Scrap: the requester states the scrap type (Priority: P2)

A work center has scrap to move. There is no job to read a scrap decision from, so the wizard asks the operator which scrap type it is. The list they choose from is the same configured list the plant already maintains, and the card tells the handler which lugger that scrap type goes to, following the lugger arrangement's own rules for a filled lugger.

Why this priority: scrap is a physical pickup that cannot be deferred; naming the wrong scrap type sends a handler to the wrong lugger.

Independent Test: raise a generic scrap request, choose each configured scrap type in turn, and confirm the card names the lugger the arrangement assigns for that type; then mark that lugger full and confirm the card follows the arrangement to the backup.

Acceptance Scenarios:

1. **Given** the scrap item is chosen on a generic request, **When** the wizard reaches the scrap step, **Then** it asks the requester to state the scrap type and reads no job.
2. **Given** the scrap type question, **Then** the choices are the plant's single scrap-type list — the same list the Setup scrap picker and the lugger arrangement editor already use — shown in that list's order, together with any type the arrangement records that is outside it.
3. **Given** a chosen scrap type, **When** the card is drawn, **Then** it names that scrap type and the lugger the arrangement assigns to it.

4. **Given** a configured scrap type whose luggers are all blocked, **When** the card is drawn, **Then** it names the scrap type and the lugger in the arrangement's order with its status beside the name, and does not draw blank.
5. **Given** a scrap type that has been removed from the arrangement after a request carrying it was raised, **When** the card is drawn, **Then** it still names the scrap type the requester stated and records the missing lugger as a configuration gap rather than inventing a lugger.
6. **Given** the requester does not choose a scrap type, **When** they try to continue, **Then** the flow does not advance.

User Story 5 - A die is found by searching Visual (Priority: P2)

A work center needs a die. The requester knows either the die's own FGT number or the part the die makes, so the wizard offers a search page that accepts either. A die number that exists in Infor Visual is accepted; searching by part number lists every die Visual ties to that part, each with the location it lives in — because a busy part usually has several dies. The requester picks the die or dies they need, and each becomes its own request, exactly as a work-order request with several dies already behaves.

Why this priority: dies are the expensive, easily-misdirected item on this list; an unverified die number sends two people to the tool crib for nothing.

Independent Test: search with a die number, then with a part number that owns several dies, and confirm every die and location is listed; then try a number that exists nowhere and confirm it is refused.

Acceptance Scenarios:

1. **Given** the die item is chosen on a generic request, **When** the die step is entered, **Then** it is a search page that accepts either a die number or a part number, and nothing is read from a job.
2. **Given** a search term, **When** it matches, **Then** the results list every matching die with its own location, or state that the die's location is not recorded.
3. **Given** a part number that several dies are associated with, **When** the search runs, **Then** all of those dies are listed together with their locations, not just one.
4. **Given** the results list, **When** the requester chooses one die, **Then** that die is the request's die.
5. **Given** the results list, **When** the requester chooses several dies, **Then** one request is raised per chosen die and the confirmation step lists them all, so the review never understates what is about to be raised.
6. **Given** a term that matches no die, or a typed die number that does not exist in Infor Visual, **When** the requester tries to continue, **Then** the flow refuses to advance and says no die matched.

7. **Given** Infor Visual cannot be reached, **When** the die search runs, **Then** the page reports the source unavailable with a retry and accepts no unverified die.

User Story 6 - A generic request still carries an honest time estimate (Priority: P2)

A handler arranging the day looks at a generic request and sees how long it should take, taken from the time already configured for that request type — not from a work order, and not from a guess. If the request type has no time configured, the request shows no estimate rather than a made-up one.

Why this priority: the estimate is what makes a request schedulable; a wrong or invented figure is worse than an empty one.

Independent Test: raise a generic request for a type with configured time and one without, and confirm the first shows the configured minutes and the second shows nothing.

Acceptance Scenarios:

1. **Given** a generic request, **When** it is shown on the board or its detail, **Then** the time it needs is the allotted time configured for its item — the same value a work-order request of that item uses.
2. **Given** an item with no allotted time configured, **When** a generic request for it is shown, **Then** no estimate appears and no default is invented.
3. **Given** the same item raised as a work-order request and as a generic request, **When** both are shown, **Then** both carry the same configured time.

User Story 7 - A generic request's card says what it is (Priority: P2)

A material handler reads the board. A generic request's card cannot show a work order, a die, a coil or a flatstock, because none of those exist for it. Instead the card identifies the request by the part number and the work center that asked for it, carries the category and item wording in the same place the other cards carry theirs, and shows scrap type, dunnage part and the needed-by time when they apply. Generic and work-order cards sit in the same list and are told apart at a glance.

Why this priority: the card is the only surface most handlers ever see, and a card built for a work order would draw empty cells where facts should be.

Independent Test: raise one generic and one work-order request, view the board, and confirm each card shows only facts that exist for it and that the generic card carries no work-order value anywhere.

Acceptance Scenarios:

1. **Given** a generic request, **When** its card is drawn, **Then** it uses the Generic Request card layout and draws no work-order, die, coil, flatstock or component cell.
2. **Given** a generic request, **When** its card is drawn, **Then** its identity is the part number and the requesting work center, and no work order number appears anywhere on it.
3. **Given** a generic request for scrap, **When** its card is drawn, **Then** the scrap type and the assigned lugger appear as they do on a work-order scrap card.
4. **Given** a generic request for dunnage, **When** its card is drawn, **Then** the chosen dunnage part appears.
5. **Given** a generic request, **When** its card is drawn, **Then** the needed-by time the requester gave appears.
6. **Given** a board holding both kinds, **When** both cards are drawn, **Then** the generic card is distinguishable from the work-order card without opening either, and neither card is drawn blank or placeholder-filled.
7. **Given** a generic request, **When** its detail page is opened, **Then** it shows the same facts as the card plus the requester, the time the request was raised, and the request's history.

User Story 8 - The generic designation is maintained, not hard-coded (Priority: P3)

A setup lead opens the work center catalog and marks the work centers that raise generic requests, and clears the mark when a work center goes back to running jobs full time. A person without that permission cannot change it. No work center is generic because a build said so.

Why this priority: the mark has to survive the plant changing its mind, and it must not be a code change.

Independent Test: as a permitted user, mark a work center generic, raise a request, clear the mark, raise another, and confirm the first used the generic flow and the second the work-order flow; then confirm an unpermitted user cannot reach the control.

Acceptance Scenarios:

1. **Given** a person holding the permission that already governs the work center catalog, **When** they open it, **Then** they can mark a work center as generic and clear that mark.

2. **Given** a work center is marked generic, **When** a request is started there, **Then** the generic flow is used.
3. **Given** the mark is cleared, **When** the next request is started there, **Then** the work-order flow is used and no generic-only step is offered.
4. **Given** a generic request is in progress, **When** the mark is cleared mid-request, **Then** the in-progress request completes as generic.
5. **Given** a person without that permission, **When** they reach any route into the catalog, **Then** they cannot see or change the generic mark — not shown-but-disabled.

Edge Cases

- A generic work center has an active job and the requester wants the job anyway: no job-derived path is offered from a generic work center; the request is generic (User Story 1, scenario 6).
- Both the WIP application and Infor Visual are unreachable: a part cannot be proved, so no request can be raised from that step; the step reports both sources unavailable with retries, and no cached part list is offered.
- A number exists in Visual and in the WIP application: the requester's WIP/finished-good answer decides which system proves it, and the other system is never consulted.
- A scrap type is renamed or removed from the arrangement after a generic request carries it: the request keeps the wording the requester stated, and the lugger cell reports the missing assignment rather than an invented lugger.
- A die has no location recorded: the result row lists the die and states that no location is recorded, rather than an empty cell.
- A die search returns more dies than fit one screen: every match is reachable, and the requester can see how many matched.
- Two requesters raise the same part at the same work center: two requests exist, because two people are asking.
- A generic request is cancelled or the app closes mid-flow: nothing reaches the queue, and no partial request is stored.
- The dunnage catalogue opens while the substitute picker is unavailable in Settings: the page reports the same unavailability rather than offering a shorter sample list.
- A requester raises a generic request for an item whose configured questions no longer exist: the flow reports the item's configuration as unavailable rather than silently skipping the questions.

Requirements

Functional Requirements

Designation of a generic work center

- **FR-001**: The work center catalog MUST carry a per-work-center designation that says whether requests raised there are generic, editable without a code change.
- **FR-002**: Only a person holding the permission that already governs the work center catalog MUST be able to set or clear that designation; for everyone else the control MUST be absent, not disabled.
- **FR-003**: No work center MUST be treated as generic because of a built-in list, a build configuration, or a default; an unmarked work center MUST always use the work-order flow.

The generic flow

- **FR-004**: Selecting a generic work center in the request wizard MUST continue the wizard without requiring an active job and without failing the work center because it has none.
- **FR-005**: A generic request MUST offer the same categories and the same items a work-order request offers, driven by the same configured request-item definitions.
- **FR-006**: Every answer a work-order request takes from the job MUST instead be asked of the requester, and no job value MUST be read, shown, or inferred during a generic request.
- **FR-007**: A generic request MUST NOT store a work order number, and MUST NOT display one on any surface.
- **FR-008**: A request raised at a generic work center MUST enter the same waitlist queue and appear on the same board as every other request.
- **FR-009**: The generic flow MUST be built as its own flow, and its views as their own projects, so that the work-order flow is not modified to accommodate it; the existing work-order flow's behaviour MUST remain unchanged.
- **FR-010**: Clearing a work center's designation MUST NOT alter requests already raised there; an in-progress generic request MUST complete as generic.

A stated part

- **FR-011**: Where an item requires a part number, the generic flow MUST ask first whether the part is WIP or a finished good, and MUST NOT accept a part number before that answer is given.
- **FR-012**: The part number MUST be entered through a completion control that offers matches as the requester types, and it MUST NOT be possible to submit a value that was not completed to a genuine part.

- **FR-013**: A WIP part MUST be proved against the WIP application's records, and a finished good MUST be proved against Infor Visual; the system the requester chose MUST be the only one consulted for that part.
- **FR-014**: A typed number that matches no genuine part MUST be refused with a message naming the source that was searched, and the flow MUST NOT advance with it.
- **FR-015**: Changing the WIP/finished-good answer MUST clear the previously entered part and require it to be proved again against the newly chosen source.
- **FR-016**: Once a part is proved, the system MUST check whether any of it is on hand and MUST report the finding and the system that answered.
- **FR-017**: Finding no stock on hand MUST warn and MUST NOT block the request.
- **FR-018**: Where the proving source cannot be reached, the flow MUST report that source as unavailable with the time of the attempt and a retry, MUST accept no unverified part, and MUST NOT substitute a cached, mirror, sample, or fabricated part list.
- **FR-019**: A generic request MUST NOT store a quantity; the request carries the part and, when none was on hand, the recorded fact that none was.

Dunnage

- **FR-020**: For an item that asks for dunnage, the generic flow's dunnage step MUST be the dunnage catalogue rendered as a page of the wizard, showing the same catalogue the substitute picker shows today.
- **FR-021**: A chosen dunnage part MUST become the request's dunnage and MUST appear on the confirmation step and the card.
- **FR-022**: A generic dunnage request MUST carry exactly one dunnage part.
- **FR-023**: Leaving the dunnage page without choosing a part MUST record no dunnage part, and MUST NOT advance past a step that requires one.
- **FR-024**: A catalogue that cannot be read MUST be reported as unavailable with a retry, and MUST NOT be replaced with a sample or fabricated list.

Scrap

- **FR-025**: For the scrap item, the generic flow MUST ask the requester to state the scrap type and MUST read no job.
- **FR-026**: The scrap-type choices MUST come from the single scrap-type list the plant already maintains for the Setup scrap picker and the lugger arrangement editor, offered in that list's order, plus any type the arrangement records that falls outside the list; no second scrap-type list MAY be introduced.
- **FR-027**: A stated scrap type MUST be carried on the confirmation step, on the card, and in the stored request.

- **FR-028**: A generic scrap card MUST name the lugger the lugger arrangement assigns to the stated type, following the arrangement's blocking and backup rules rather than reading a job.
- **FR-029**: A scrap type with no lugger assigned MUST be shown with the type named and the missing assignment recorded as a configuration gap, never with an invented lugger.
- **FR-030**: A scrap type removed from the arrangement after a request carries it MUST leave that request's stated type intact.
- **FR-031**: The flow MUST NOT advance without a scrap type where the item requires one.

Die

- **FR-032**: For the die item, the generic flow MUST present a die search page that accepts a die (FGT) number or a part number and reads no job.
- **FR-033**: A die number MUST be accepted only when it exists in Infor Visual, and a search by part number MUST list every die Visual associates with that part.
- **FR-034**: Each die in the results MUST be listed with its location, and a die with no recorded location MUST say so rather than showing an empty cell.
- **FR-035**: The results MUST list every matched die, including every die belonging to a part that has several, with a visible count of how many matched.
- **FR-036**: Choosing several dies MUST raise one request per chosen die, and the confirmation step MUST list every one of them.
- **FR-037**: A term matching no die, or a typed die number that does not exist in Visual, MUST be refused with a message naming the source searched, and the flow MUST NOT advance.
- **FR-038**: Where Infor Visual cannot be reached, the die search MUST report it unavailable with a retry and accept no unverified die.

Time, urgency and description

- **FR-039**: A generic request's time estimate MUST be the allotted time configured for its item, identical to the value a work-order request of that item uses.
- **FR-040**: An item with no allotted time configured MUST produce no estimate rather than a default.
- **FR-041**: The generic flow MUST ask the requester for a needed-by time or urgency, and the answer MUST appear on the confirmation step, the card and the stored request.
- **FR-042**: The generic flow MUST allow a free-text description, and MUST NOT require one.

Confirmation, queue and record

- **FR-043**: The confirmation step MUST list everything that will be raised — every die entry, the stated part and its type, the dunnage part, the scrap type, the needed-by time and the description — so the review cannot understate the submission.

- **FR-044**: Submitting MUST write the request through the same queue path every other request uses, in one operation, so that a partial request is never left behind.
- **FR-045**: A generic request MUST record its requester, the work center, the item, the answers given, the time it was raised, and every warning the requester saw and accepted.
- **FR-046**: The requester's identity MUST come from the signed-in session and the existing employee-verification rules; the generic flow MUST NOT invent or accept a typed identity.
- **FR-047**: Fulfilment of a generic request MUST use the existing handler flow; a generic request MUST NOT need a separate handling path to be worked.

Cards and surfaces

- **FR-048**: A generic request MUST draw its own card layout, and that layout MUST NOT draw work-order, die, coil, flatstock or component cells.
- **FR-049**: A generic card's identity MUST be the stated part number and the requesting work center.
- **FR-050**: A generic card MUST show the category and item wording, the needed-by time, and, where they apply, the scrap type with its assigned lugger and the chosen dunnage part.
- **FR-051**: Generic and work-order cards MUST appear in the same list and MUST be distinguishable without opening either.
- **FR-052**: No generic card MUST draw an empty or placeholder cell in place of a fact the request does not have.
- **FR-053**: A generic request's detail page MUST show the same facts as its card plus the requester, the time raised, the accepted warnings, and the request's history.

Truthfulness

- **FR-054**: Every value a generic request shows MUST come from a read of the system that owns it, from the requester's own answer, or from configuration; no cell MAY be filled with a sample, a cached substitute for an internal store, or a guess.
- **FR-055**: Every failure to establish a required value MUST be reported on the step where it matters, with the time of the attempt and a retry, and MUST NOT be turned into a persistent banner or a silent blank.

Key Entities

- **Generic work center designation**: whether requests raised at a named work center are generic; held against the work center record itself, editable by a permitted person.
- **Stated part**: the part number the requester named, whether it is WIP or a finished good, which system proved it, and whether stock was on hand when the request was raised.
- **Dunnage selection**: the single dunnage part chosen from the catalogue page.

- **Stated scrap type**: the scrap type the requester named, with the lugger the arrangement assigns at the time the request is read.
- **Die selection**: one or more dies found by search, each with its own number and location, each producing its own request.
- **Generic request**: a queued request carrying no work order, identified by its stated part and requesting work center, with its item, answers, needed-by time, description, requester, time raised and accepted warnings.
- **Generic request card**: the board presentation of a generic request, drawing only the facts that request has.

Success Criteria

Measurable Outcomes

- **SC-001**: A requester at a generic work center can raise a complete request without a work order, in under 90 seconds for a part request and under 2 minutes for a dunnage, scrap or die request.
- **SC-002**: Zero generic requests are accepted carrying a part number, dunnage part, scrap type or die that could not be proved against its owning system.
- **SC-003**: Zero generic requests carry a work order number, on any surface, in any export.
- **SC-004**: Zero generic cards or detail pages show a sample, placeholder, or invented value in place of a fact the request does not have.
- **SC-005**: A handler can tell a generic request from a work-order request on the board without opening either, and can act on it through the existing fulfilment flow.
- **SC-006**: A part with no stock on hand is still raisable, with the warning visible on the request afterwards, in 100% of cases.
- **SC-007**: The work-order flow's behaviour is unchanged: every existing request scenario still passes with the generic flow present.
- **SC-008**: A part number that several dies belong to returns every die and location in one search, with no die silently omitted.

Assumptions

- **A1**: The generic designation is a field on the existing work center catalog, maintained through the setup/settings surface that already governs that catalog, under its existing permission.
- **A2**: The requester's identity continues to come from sign-in plus the existing employee verification; the generic flow adds no second identity step.
- **A3**: Generic requests share the existing waitlist queue and board; only their card layout differs.

- **A4**: A generic request captures no quantity; the request names the part and, when none was on hand, records that.
- **A5**: Choosing several dies raises one request per die, matching how a work-order request with several dies already behaves.
- **A6**: One dunnage part per request is the scope; several dunnage parts on one request is out of scope.
- **A7**: The scrap types offered are the plant's one existing scrap-type list, the list the Setup scrap picker and the lugger arrangement editor already share; the generic flow introduces no second list.
- **A8**: Categories, items and their configured questions are shared with the work-order flow, so a change to a request item applies to both.
- **A9**: A generic work center that still has an active job still uses the generic flow; the job is never read there.
- **A10**: A die's association with a part is established from Infor Visual, preferring an association Visual records explicitly and falling back to the part's work-order history; the outcome FR-033 requires is that every die Visual ties to the part is listed.

Open Questions

- **Q1**: Should a generic work center ever be able to read an active job on purpose, or is the job permanently out of scope once the designation is set? (Assumed out of scope, A9.)
- **Q2**: When an item names a part but the plant also needs to know where it should go, is the requesting work center always the destination? (Assumed yes; no delivery-point question is asked.)